GEORGE MASON UNIVERSITY

# CHALLENGE X

Unissant™

# Use Case Report

**Project Title:** ***Investigative AI Training Arena: Multi-Agent Interview Coaching and Evaluation***

# Problem Overview

- Investigators need repeated practice adjusting their questions when answers are incomplete or conflicting.
- Today’s training scenarios are static. They offer little variety and make it hard to compare investigative AI assistants fairly.
- Training must reward catching deceptive subjects while protecting truthful people from false accusations.
- Truthful people can be confused, stressed, or unable to recall details. Those traits, like nervousness, demeanor, or accent, must never be shortcuts to a deception label.



# Challenge

Build an AI training arena where investigators practice interviews with simulated subjects while competing AI assistants coach them. Your solution should:

- ❑ Help investigators ask better follow-up questions, using the evidence in each answer
- ❑ Detect evasive answers and identify which subjects were assigned a perpetrator role
- ❑ Compare competing AI assistants fairly against truthful and deceptive subject agents

Goal:

- ❑ Improve investigators' questioning and evidence-backed findings through repeatable AI agent competitions



# Proposed Solution

Build a text-based training platform with two types of competing AI agents:

- ❑ Investigative assistants: Review each question and answer, flag gaps and contradictions with transcript evidence, and suggest a neutral follow-up
- ❑ Simulated subjects: Play a truthful or deceptive role from a private profile; deceptive subjects try to hide an assigned violation within the scenario rules
- ❑ Scenario engine: Locks case facts, releases new evidence on a script, sets interview limits, and hides role labels and scoring keys
- ❑ Independent scoring: Compares each final finding (deceptive, truthful, or not enough evidence) with the hidden case facts
- ❑ Two modes: Practice gives live coaching; assessment saves feedback for the debrief



# Agent Roles

## Investigative assistants

- ❑ Assess each question against the answer received.
- ❑ Flag gaps and contradictions with transcript evidence.
- ❑ Suggest a neutral follow-up and a finding with confidence.

Goal: Supported findings with few false accusations

## Simulated subjects

- ❑ Receive a private scenario profile and assigned role.
- ❑ Truthful roles answer from known facts and may be uncertain.
- ❑ Deceptive roles try to conceal the assigned violation within scenario rules.

Goal: Profile fidelity plus the assigned truthful or evasive objective



# Scenario Design

- ❑ Assign a role, event timeline, permitted knowledge and a private objective
- ❑ Mix deceptive and truthful cases across matched levels of difficulty
- ❑ Vary communication style independently of the assigned role
- ❑ Lock case facts for each trial. Release new evidence through scripted events
- ❑ Keep role labels and scoring keys hidden from investigators and assistants

*Dynamic behavior must remain consistent with the scenario's ground truth*



# Agent Bake-Off

Proposed MVP
3 assistant and 3 subject candidates

**01**

Matched cases
Cross-pair assistant and subject candidates under equal budgets.

**02**

Two test tracks
Replay identical transcripts, then run interactive interviews.

**03**

Blinded scoring
Compare locked findings with case facts and trainer ratings.



# Example Output

***Ground rule:*** *All cases are synthetic. Evasive wording alone does not prove a violation, and a simulated score is not a finding about a real person.*

**Challenge test bed:** Run a bake-off on matched synthetic cases with 3 assistant and 3 subject candidates under equal budgets. Replay identical transcripts, then run live interviews.

**01**

**Interview exchange**

The investigator asks, “Why did you enter?” The subject replies, “It was a busy shift.”

**02**

**Assistant flag**

The purpose of entry is unanswered. Suggested neutral follow-up: “What task required entry?”

**03**

**Evidence-based finding**

More evidence needed. Verify the work order and timeline before reaching a finding.



# Evaluation

Separate scorecards for learning, detection and subject realism.

- Investigators: question quality, evidence coverage and improvement on unseen cases.
- Assistants: evasion-event precision/recall, role recall and false-positive rate.
- Subjects: consistency with assigned facts and success within scenario rules.
- All trials: unsupported flags, unresolved findings, latency and cost.

Assistant ranking: pass false-positive and isolation gates, then compare recall and questioning quality.



# Impact & Goals

- ❑ Improve investigators' questioning score by at least 20% over their own starting baseline
- ❑ Identify at least 80% of deceptive roles; misclassify no more than 5% of truthful roles
- ❑ Keep simulated subjects in character: at least 95% profile fidelity
- ❑ Deliver coaching within 3 seconds for 95% of interview turns
- ❑ Rank fairly: assistants must first pass the false-accusation and isolation (no access to hidden answers) checks, then compete on detection and questioning quality



# Prize

Possibility for internship opportunities at Unissant